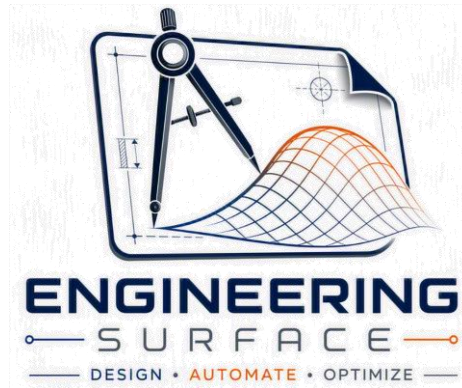




Engineering Surface

CATIA Stack-Up Dashboard

[User Guide & Help Documentation](#)

Version 3.7.5 • AURORA Professional Edition

Professional tolerance stack-up analysis for CATIA assemblies

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1. Introduction

CATIA Stack-Up Dashboard is a Windows tool that performs tolerance stack-up analysis on CATIA assemblies. It lets you:

- Define stack-ups: a chain of dimensions (contributors) that build up to a closing dimension (the Gap).
- Import contributor names directly from CATIA by picking two faces.
- Compute the gap with Worst-Case, RSS (Root Sum Square) and Monte-Carlo methods.
- Compare the result against a specification (Target $\pm$ tolerance) and get Cpk, Yield %, failures per million and a clear PASS / MARGINAL / NOT OK verdict.
- Generate a professional Excel report with one sheet per stack-up, including CAD section screenshots and charts.

Key concepts

Term	Meaning
Stack-Up	One analysis chain. Example: "door gap", "cover to housing clearance".
Contributor	One dimension of the chain: Nominal value, tolerances IT(+) / IT(-), and a sign (+ adds, - subtracts).
Gap (closing dimension)	The RESULT — never measured or picked; computed from all contributors.
Specification	Your requirement: a Target value plus upper (+) and lower (-) tolerances → defines LSL and USL.
Worst Case	Min/max gap if every contributor is simultaneously at its extreme tolerance.
RSS	Statistical combination assuming independent, normally-distributed contributors.
Cpk	Process capability index of the computed chain vs. the specification.
Yield	% of produced assemblies expected to fall inside the specification (Monte-Carlo).

2. System Requirements & Installation

Item	Requirement
Operating system	Windows 10 / 11 (64-bit)
Framework	.NET Framework 4.8 (installed automatically with the tool if missing)
CATIA	Any version exposing the CATIA COM interface (V5/V6 via Automation); an open CATPart or CATProduct must be active
Excel	Microsoft Excel 2010 or newer — required for the Excel report export
Disk / RAM	~50 MB disk, 2 GB RAM minimum

Installation:

- Run CATIA_StackUp_Professional_Dashboard.exe — no installer needed (portable).
- First launch creates a 14-day free trial automatically.
- To make it available for all users, place the exe in a shared folder (e.g. C:\Program Files\CATIAStackUp). Per-user data is stored in %APPDATA%\CatiaStackUp.

Antivirus note: some antivirus software may flag the exe because it automates CATIA and Excel. Add an exclusion if this happens.

3. Licensing & Editions

The tool runs as a 14-day free trial after first launch. During the trial, some features are locked. To continue beyond the trial, purchase a license from your vendor.

Editions & prices

Feature	Free Trial	Standard	Professional	Company
Price	—	\$300 / year or \$50 / month	\$500 / year or \$80 / month	\$1500 / year or \$120 / month
Stack-ups calculated	2	5	Unlimited	Unlimited
Contributors per stack-up	3	20	Unlimited	Unlimited
Excel export	Locked	Yes (no company logo)	Yes + company logo	Yes + company logo
PASS spec quick-set	Locked	Yes	Yes	Yes
Analysis charts	Locked	Yes	Yes	Yes
Change the company logo	Locked	Yes	Yes	Yes
Machine-bound key	—	Optional	Optional	—
Seats on the key	—	1	1	1–250

How to activate your license

1. Open the tool and click the orange Buy License button (lock icon, top-right).
2. The License & Purchase dialog shows the three editions with prices and your Machine code (e.g. PC123 - john.doe).
3. Send your name + Machine code to your vendor and tell them which edition you want.
4. The vendor sends you an activation key (format: XXXXXX-XXXXXX-XXXXXX-XXXXXX).
5. Paste the key into the dialog, enter your name, click Activate.
6. You get a confirmation: "License activated successfully! Edition: ... Valid until: ..." — all features unlock immediately, no restart needed.
7. The Buy License button turns green and reads Licensed; the status bar shows your edition and expiry date.

License & Purchase

CATIA Stack-Up Dashboard - Licensing

Your current license

No active license - you are using the FREE TRIAL.
Trial limits: 2 stack-ups, 3 contributors,
Excel export, PASS spec and charts locked.
Choose an edition below to purchase.

3 license editions are available - pick the one that fits your needs:

Editions prices

Standard - \$300 / year or \$50 / month
5 stack-ups, 20 contributors, Excel export

Professional - \$500 / year or \$80 / month
Unlimited stack-ups contributors, Excel export + company logo

Company - \$1500 / year or \$120 / month
Multi-seat license (up to 250 users), everything included

Price details purchase, contact: ennadisto@gmail.com

After purchase, paste the activation key below:

Machine code: HOME - ENNAD

Activation key:

Customer:

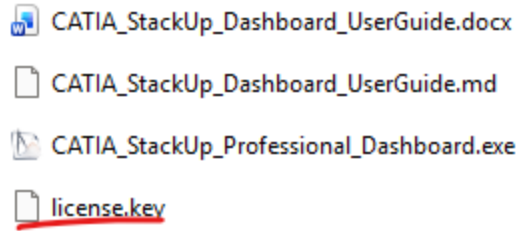
Send the Machine code above + your name to: ennadisto@gmail.com
A machine-bound key only works on this PC.

Activate

Close

Important notes about keys

- Machine-bound keys only work on the one PC whose machine code was used. Ask your vendor for a new key if you get a new PC.
- Expiry: when a license expires, the tool falls back to trial limits automatically. Renew by getting a new key with a later validity date.
- Upgrade / downgrade: simply activate the new key — the new edition's limits apply immediately.
- The license is stored in a file named license.key next to the exe. **Do not delete it.**



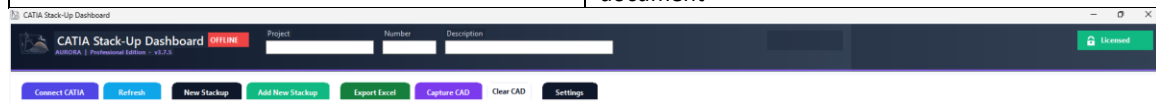
4. Getting Started — First Launch

1. Start CATIA and open the assembly (.CATProduct) or part you want to analyze.
2. Start the CATIA Stack-Up Dashboard.
3. Click Connect CATIA in the command bar. The status pill turns CONNECTED (green), the status bar shows the document name, and the left panel fills with the CATIA product structure.
4. Fill in the project fields in the header: Project, Number, Description (they appear in the Excel report).

If the connection fails, see Troubleshooting (section 16).

5. The Interface — A Guided Tour

Zone	What it does
Header	Company logo (owner default), title + version, CATIA connection pill, project fields, Buy License button
Command bar	Connect CATIA, Refresh tree, New Stackup, Add New Stackup, Export Excel, Capture CAD, Clear CAD, Settings, Buy License
Left panel	CATIA product structure (read-only browser of the active assembly)
Center panel	Stack-up selector + Gap requirement, contributor toolbar, KPI cards, contributors grid, result bar
Right panel	Contributor editor (top) + Analysis Charts tabs (Contribution %, Cpk, Stack Diagram, Status/Recs, Top Contrib.)
Status bar	Live messages, trial/license status, connected document



6. Core Workflow — Step by Step

Step 1 — Create a stack-up

- On first launch a stack-up named "Stackup 1" exists. Use Add New Stackup to create more (one gap per stack-up).
- Use New Stackup to clear everything and start over (it deletes all stack-ups).

- Select a stack-up in the Stackup dropdown to switch; Delete removes it.

Step 2 — Add contributors (manually or from CATIA)

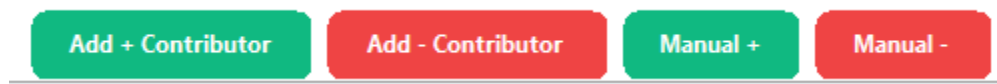
Option A — Manually (no CATIA needed):

1. Click Manual + or Manual – in the contributor toolbar.
2. A new contributor row appears; the editor opens so you can type the Name, Nominal 1, IT1(+), IT1(–) yourself.
3. Click Apply to row.

Option B — From CATIA (imports part names):

1. Click Add + Contributor (positive direction) or Add – Contributor (negative direction).
2. CATIA comes to the front; a message asks you to select two FACES.
3. Click the first face in the 3D view, then the second face.
4. Back in the tool, a new contributor row appears with the two part names imported.

Only the names are imported. You type the Nominal and tolerances yourself.



Step 3 — Enter the contributor values

Field	Description
Name	Contributor name (default: the two picked part names)
Part 1 / Part No / Rev	Identification (imported / editable)
Nominal 1	The nominal dimension (current units). Type a NEGATIVE number to make the contributor negative.
IT1 (+)	Upper tolerance (positive number)
IT1 (–)	Lower tolerance (positive number)
Sign	+ or – (direction in the chain)
Datum	Optional reference (documented in the report, not used in math)
Proc. σ	Optional real process standard deviation (mm). If empty, estimated from tolerances: $\sigma = (IT+ + IT-) / 2 \div \text{sigma level}$
Description	Free text, exported to the report

Click Apply to row (or finish editing the cell) — the results update automatically. You can also edit cells directly in the grid; Move Up / Move Down reorders the chain; Remove deletes a contributor.

Contributor: Contributor 1

Name:

Part 1:

Part No: Rev:

Nominal: IT1 (+): IT1 (-):

Sign:

Datum: Proc. s:

Description:

Description:

Apply to row

The Gap is the RESULT of all contributors. Enter Nominal / IT for each contributor - the Gap updates automatically.

Step 4 — Define the Gap requirement (specification)

In the center panel: Gap: [Target] [+] [-]. Target = the nominal gap you want (e.g. 0.5 mm); + → USL = Target + (+); - → LSL = Target - (-).

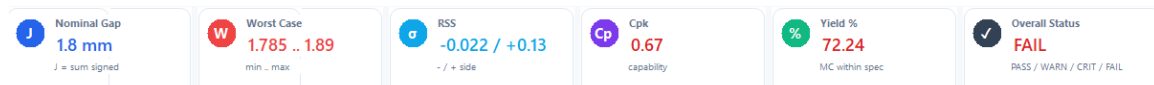
Two helpers:

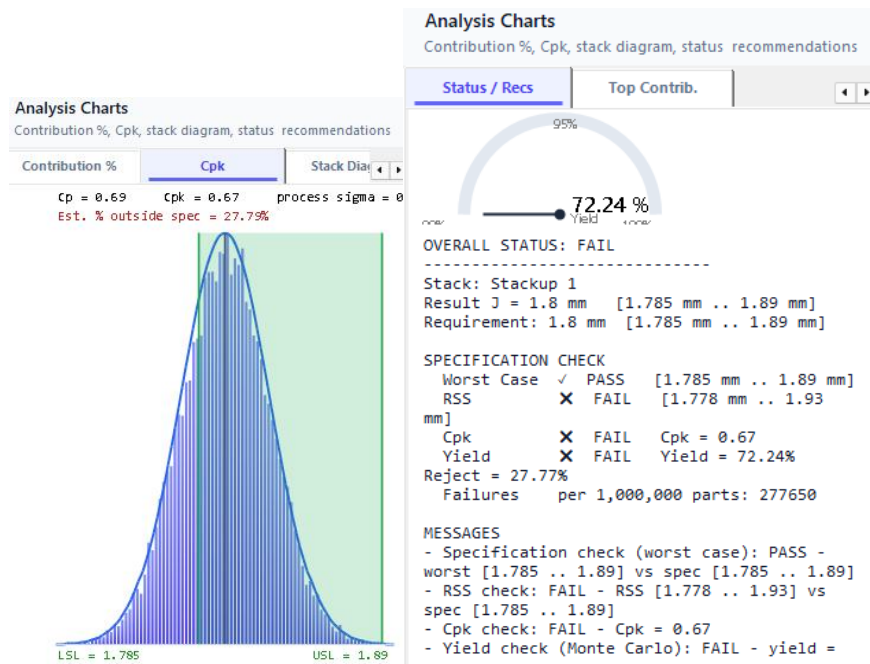
- Use J — fills Target with the computed gap J and +/- with the computed gap IT. Use it when you don't know the target yet.
- PASS spec — sets Target = computed J and +/- = the minimum width that passes with Cpk ≥ 1.67.

The gap itself is a result, never typed. Only the requirement is typed.

Step 5 — Read the results

- KPI cards: Nominal Gap (J), Worst Case (min..max), RSS (-/+), Cpk, Yield %, Overall Status (traffic-light colors).
- Result bar (bottom of the grid): full summary — J, [min..max], IT(+)/IT(-), links & method, RSS, spec, Cpk, status, failures per million.
- Analysis Charts (right panel): see section 9.





7. Understanding the Calculations

7.1 Gap nominal (J)

The closing dimension is the sum of the signed contributor nominals (thermal expansion applied if $dT \neq 0$):

$$J = \sum (\text{sign}_i \times \text{Nominal}_i)$$

7.2 Worst case

$$\text{Gap min} = J - \text{IT}(-)\text{total} \quad \text{Gap max} = J + \text{IT}(+)\text{total}$$

7.3 The maillons (links) rule

- ≤ 5 contributors (links): quadratic method — Gap IT = arithmetic sum $\div 2$.
- > 5 contributors: arithmetic method — Gap IT = plain sum.
- The applied method is shown in the result bar: "Links: N (Quadratic (links ≤ 5))".

7.4 RSS (Root Sum Square)

$$\begin{aligned} \text{RSS}(+) &= \sqrt{\sum \text{IT}(+)_i^2} \times \text{sigmaFactor} & \text{sigmaFactor} &= 1 (\sigma=3) \text{ or } 2 (\sigma=6) \\ \text{RSS}(-) &= \sqrt{\sum \text{IT}(-)_i^2} \times \text{sigmaFactor} \end{aligned}$$

7.5 Process sigma of the chain

$$\sigma_{\text{chain}} = \sqrt{\sum \sigma_i^2}$$

7.6 Cpk

$$\text{Cpk} = \min((USL - J) / (3\sigma), (J - LSL) / (3\sigma))$$

Status thresholds: OK if $\text{Cpk} \geq 1.33$ and worst case inside spec; MARGINAL if 1.00–1.33; otherwise NOT OK.

7.7 Monte-Carlo & Yield

- 20,000 simulations draw each contributor from a normal distribution (mean = nominal, σ = per contributor).
- Yield % = share of simulated gaps inside [LSL, USL].
- Failures per 1,000,000 = $(100 - \text{Yield\%}) / 100 \times 1,000,000$.

7.8 Overall status logic

FAIL (red) → CRITICAL (orange) → WARNING (orange) → MARGINAL (yellow) → PASS (green), based on worst-case check, RSS check, Cpk and yield. The Status / Recs tab explains why and gives recommendations.

8. The Gap Requirement (Specification)

- A requirement is "defined" as soon as you type in any of the three fields (Target, +, -).
- Cpk / Yield / Status need a non-zero spec width — if USL = LSL you'll see "SPEC WIDTH IS ZERO".
- Use Use J to avoid guessing the target.
- Use PASS spec to get the minimum width that passes ($\text{Cpk} \geq 1.67$).

9. Analysis Charts

Available in the Analysis Charts card (right panel); locked in the free trial.

Tab	Shows
Contribution %	Sensitivity: each contributor's share of the total tolerance (blue: % of IT width; orange: % of IT ²). The biggest bar = the driver of your stack.
Cpk	Distribution of the simulated gap (blue histogram + curve), green specification band [LSL..USL], Cp/Cpk values and estimated % outside spec.
Stack Diagram	Visual chain: blocks for + contributors (right of 0) and - contributors (left), with the gap bracket.
Status / Recs	Verdict with icons (✓ PASS, ⚠ WARNING/MARGINAL, ✗ FAIL), all checks, messages and auto-generated recommendations.
Top Contrib.	Top 10 contributors ranked by tolerance share, color-coded by severity.

10. Settings

Setting	What it does
Materials	List of materials with thermal expansion coefficient (1/°C). Defaults: Metal 0.000012, Plastic 0.000125. Add/remove your own.
Material for all parts	Which material's coefficient is applied to every component.
dT (ΔT)	Temperature change in °C. Effective length = $L \times (1 + \alpha \times dT)$. Set 0 to ignore thermal effects.
Sigma level	3 or 6. Scales the RSS result ($\times 1$ at $\sigma=3$, $\times 2$ at $\sigma=6$) and the yield estimates.
Units	mm or inch.
Method	Worst + RSS (full), Worst Case only, RSS only.
Company logo	Choose / remove your own logo (locked in the free trial — see section 13).

Settings

Materials (coefficient is used for thermal expansion):

Material	Coefficient (1/degC)
Metal	0.000012
Plastic	0.000125

Material for all parts:

Metal

Temperature change dT (deg C):

0

 $L_{eff} = L \times (1 + \alpha \times dT)$

Sigma level:

3

RSS is scaled by sigma/3

Units:

mm

Method:

Worst + RSS

Company logo (app header + Excel reports):

Choose Logo...

Remove

No custom logo (default used).

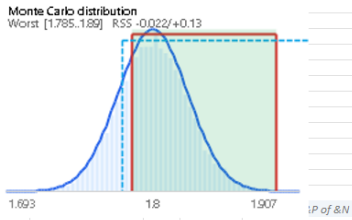
OK

Cancel

[illegible]

GRAPHS

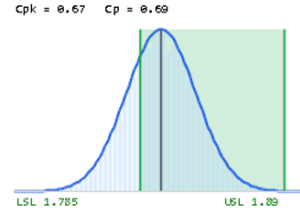
Monte Carlo distribution



Tolerance contribution



Cpk - process vs spec



Stackup 2

Tolerance Stack-Up Analysis



Dim List

Date	Name	Rev	Status
08.07.2026	XXX	1	MARGINAL

Date: 08/07/2026 Project:

Number:

Description:

Units:

Method:

Worst + RSS

Material:

Metal (alpha=0.00012)

dT:

0 degC

Sigma:

3

STACKUP 2: Stackup 2

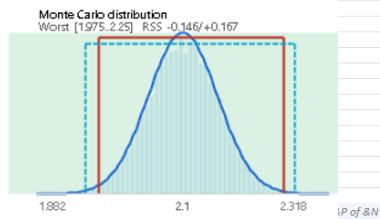
Part Number	Part Name	Rev	Dim Tag	Dimension Description	- or +	Nominal Value	Tol Type	Tol	Bilateral -	Bilateral +	Cp	Contribution %	Dimension Min	Dimension Max
RRT30			Q10		+	35.3000	Symmetric	0.1000	0.1000	0.1000		36.4%	35.2000	35.4000
RT33			Q03		-	-19.0000	Bilateral	0.0200	0.0200	0.0500		12.7%	-19.0500	-18.9800
RT45			Q25		-	-10.5000	Bilateral	0.1000	0.1000	0.1200		40.0%	-10.6200	-10.4000
RT39			Q20		-	-3.7000	Symmetric	0.0300	0.0300	0.0300		10.9%	-3.7300	-3.6700

Notes / Diagrams

Nominal Stack	2.1 MM
Shifted Nominal Stack	2.1
Worst Case +/-	1.975 2.25
	0.125 0.15
3 sigma RSS +/-	1.954 2.267
	0.146 0.167
Summary	
OVERALL STATUS: MARGINAL	
Overall sigma:	3
Failures per:	1.000.000
Dimensional Targets	Min 1.8400 Max 2.3600
Assemblies Failures	Left 0.3 Right 0.3
Total Failures	1

GRAPHS

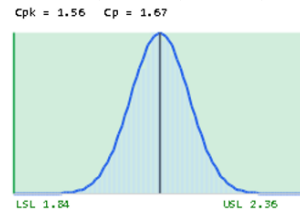
Monte Carlo distribution



Tolerance contribution



Cpk - process vs spec



12. CAD Screenshots per Stack-Up

1. Select the stack-up in the Stackup dropdown.
2. Adjust the view in CATIA (zoom/rotate/section plane) exactly as you want it in the report.
3. Click Capture CAD → the message names the stack-up → OK → the screenshot is taken, cleaned (tree/compass removed, background transparent) and attached to that stack-up only.
4. The Capture CAD button shows ✓ when the current stack-up has a screenshot; hover it to see the path.
5. Clear CAD removes the screenshot from the current stack-up only.

The screenshot appears in the stack-up's own Excel sheet under "CAD Section View – <stack-up name>".

13. Logo & Branding

- Default (owner) logo: the tool ships with app_logo.png next to the exe; it is shown in the app header and in the Excel reports.
- User logo: licensed users can set their own logo in Settings → Company logo (Choose Logo... / Remove). It is stored per-user in %APPDATA%\CatiaStackUp\logo.png and overrides the default for that user only.
- Free trial: changing the logo is locked — trial users always see the owner's default logo.

14. RSS Demonstration Dialog

The RSS Demo button opens a step-by-step explanation of the RSS calculation: each contributor's $IT(+)$ / $IT(-)$, their squares, the sum of squares, the square root (base RSS), the sigma factor ($\sigma/3$), and the final RSS(+/-) with the gap sign swap. Useful for training and documentation.

15. Frequently Asked Questions

Q1 — Why is the gap "0" or not computed?

The gap is the result of all contributors. Add at least one contributor and enter Nominal/IT values. With no values, the result bar says PENDING.

Q2 — My Cpk chart is empty.

The Cpk chart needs tolerance data: enter IT(+) and IT(-) for each contributor (or a process sigma). With all-zero tolerances the chart shows a message instead of drawing.

Q3 — Can I use the tool without CATIA?

Yes — completely. Use the Manual + / Manual – buttons in the contributor toolbar to add contributors by typing Name, Nominal and tolerances yourself. CATIA is only needed for face picking (Add + / Add –) and for CAD screenshots (Capture CAD).

Q4 — What happens when my trial ends?

The tool shows a warning and opens the License dialog. Until you activate a key, it keeps working with trial limits (2 stack-ups, 3 contributors, locked export/charts/logo).

Q5 — Can I reinstall to get a fresh trial?

No. The trial start date is stored per Windows user in %APPDATA%\CatiaStackUp (and next to the exe); reinstalling to another folder does not reset it.

Q6 — I bought a license, why is it still locked?

The key must be activated (pasted in the Buy License dialog), not just copied next to the exe. If you placed license.key manually, restart the tool. Check the status bar for the license message.

Q7 — Does the tool work offline?

Yes — everything runs locally. License activation is offline (key-based).

Q8 — Can several people share one license?

A Company license supports multiple seats (1–250) via its key. Standard/Professional keys are single-user; if the vendor enabled machine binding, the key is tied to one PC.

16. Troubleshooting

CATIA connection fails

- Start CATIA first and open a CATPart/CATProduct.
- Run the tool as the same Windows user as CATIA.
- Use the x64 build (default) if your CATIA is 64-bit.
- Close extra CATIA sessions (multiple CNEXT processes confuse the connection).
- Click Refresh after connecting to reload the product tree.

Excel export fails

- Make sure Excel is installed.
- The tool offers a CSV fallback — accept it to still get the data.

Screenshot problems

- Capture per stack-up: select the stack-up first.
- If the capture looks wrong, adjust the CATIA view and re-capture (the old image is replaced).

Trial / license messages

- "Trial expired" → contact your vendor to purchase/renew.
- "Invalid activation key" → check the key format (24 characters, 4×6 groups) and that a machine-bound key matches this PC's machine code (shown in the Buy License dialog).

The tool won't start

- Install .NET Framework 4.8 (Windows will offer it).
- Check antivirus exclusions for the exe folder.

17. Support & Purchasing

- Buy: click Buy License in the tool — see section 3 for editions & prices.

Price details & purchase, contact: ennadisto@gmail.com

- Machine code: always available in the Buy License dialog — send it with your order.
- Support: provide your edition, valid-until date (status bar) and the exact text of any error message.

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